Batuhan Bayır

Born: 14.09.1999, İzmir/Türkiye
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Education

3. **The University of Utah** (*Aug. 2023 — present*) — Ph.D. student in Mathematics
    - Advisors: Prof. Yekaterina Epshteyn and Prof. William M Feldman
2. **Ozyegin University** (*Sept. 2021 — July 2023*) — M.Sc. in Mathematics — CGPA is *3.90* over *4.00*
    - Thesis Title: Analysis of a fully discrete Fourier pseudospectral method for the Rosenau equation
1. **Ankara University** (*Sept. 2017 — June 2021*) — B.Sc. in Physics — CGPA is *3.63* over *4.00* — *Rank*: 1st — *262/240* ECTS
    - Courses taken from the math dept. : Theory of Groups, Theory of Modules, Category Theory, Real Analysis Integral Equations, Differential Geometry, Differentiable Manifolds, Lie Groups, and History of Mathematics.

Publications

1. Batuhan Bayir, Yekaterina Epshteyn and William M Feldman, JKO-Based Framework for a Nonlinear Fokker-Planck Type Model of Grain Growth, *Submitted*, 2026, View the PDF.
2. Batuhan Bayir, Yekaterina Epshteyn and William M Feldman, Global well-posedness of a nonlinear Fokker-Planck type model of grain growth, *Discrete and Continuous Dynamical Systems (DCDS)*, 2026, 48: 404-422. doi: 10.3934/dcds.2025146
3. Bootstrap 3 for Beginners (Yeni Başlayanlar İçin Bootstrap 3) — January 2015
    - Bootstrap 3 is modern CSS framework for developing mobile device friendly websites. I wrote Türkiye's first Bootstrap 3 book when I was a 15 years old. Published by KODLAB in 2015. The book comes with a supplementary DVD. Sample video on YouTube: (click).
      First Edition: Jan. 2015, Second Edition: Nov. 2015. ISBN (Softcover): 9786059118026.
      Sample pages in Google Books: (click).
4. Hermitian Matrices as a Complex Vector Space (unpublished note) — July 2020
    - In this short note, I prove that there exists a *complex* vector space structure on a set of Hermitian matrices via the *Axiom of Choice*. ResearchGate page of the note: (click).

Poster & Talks

1. JKO-Based Framework for a Nonlinear Fokker-Planck Type Model of Grain Growth
   Presented at:
    - NSF RTG Conference: Materials and Optimization (Talk), 16-18 October 2026, University of Utah

- Frontiers of Mathematical and Computational Aspects of Materials Science (Poster), 12-14 October 2026, Carnegie Mellon University
- Applied Math Collective (Talk), 22 September 2026, University of Utah

Slides: (click) and Poster: (click).

2. Global Well-Posedness of a Nonlinear Fokker-Planck Type Model of Grain Growth
Presented at:

   - RMMC Summer School (Poster), 17-20 June 2025, University of Wyoming
   - Frontiers in Applied Analysis (Poster), 3-6 June 2025, Carnegie Mellon University
   - 2025 SIAM Wasatch Student Chapters Conference (Talk), 12 April 2025, Utah State University
   - Applied Math Collective (Talk), 7 April 2025, University of Utah

   Slides: (click) and Poster: (click).

3. A Fourier Spectral Method for the Rosenau Equation — Horizons in non-linear PDEs Summer School, 26-30 September 2022, Ulm University, Germany

   - I presented the first few results of my master's thesis in the poster session of summer school. In my thesis, I propose a numerical scheme for the *Rosenau equation* and do a convergence and stability analysis of the proposed scheme. Poster: (click).

4. Tautochrone Curve and Integral Equations — 9th Bahar Mathematics Meeting, 1-2 May 2021, Zoom

   - I talked about integral equations, Laplace transform and Tautochrone curve.
   Video of talk: (click) and slide of talk: (click).

5. Mathematics of Minkowski Spacetime — 6th Bahar Mathematics Meeting, 19-20 October 2019, Hacettepe University *and* Ankara University MathCom Society Workshop, 10-11 July 2021, Zoom

   - I started with theory of bilinear forms, then I defined the Minkowski spacetime, and I proved some interesting geometric results on this space. Video of talk: (click) and slide of talk: (click).

6. Physical Aspects of Lie Theory — 5th Bahar Mathematics Meeting, 2-3 March 2019, İstanbul Bilgi University

   - I started with defining some concepts such as group and manifold, then I talked about how Lie groups and Lie algebras appeared in classical and quantum physics.

7. A Brief Introduction to Lagrangian Mechanics — 4th Bahar Mathematics Meeting, 13-14 October 2018, Middle East Technical University

   - I talked about variational problems and fundamentals of Lagrangian mechanics.

Attended Schools & Conferences

1. NSF RTG Conference: Materials and Optimization, 16-18 October 2026, University of Utah
2. Frontiers of Mathematical and Computational Aspects of Materials Science, 12-14 October 2026, Carnegie Mellon University
3. Partial Differential Equations for Many Particles Systems, 3-7 August 2026, ICERM

4. 2026 SIAM Wasatch Student Chapters Conference, 11 April 2026, University of Utah
    - Assisted in organizing the conference.
5. RMMC Summer School, 17-20 June 2025, University of Wyoming
6. Frontiers in Applied Analysis, 3-6 June 2025, Carnegie Mellon University
7. NSF CompMath Meeting 2025, 8-9 May 2025, University of Utah
    - Assisted with front-desk registration and provided technical support during the meeting.
8. 2025 SIAM Wasatch Student Chapters Conference, 12 April 2025, Utah State University
9. Horizons in non-linear PDEs Summer School, 26-30 September 2022, Ulm University, Germany
10. Conference on Mathematics of Wave Phenomena, 14-18 February 2022, *Online*
11. Computation, Analysis and Applications of PDEs with Nonlocal and Singular Operators, 4-11 February 2022, *Online*
12. Turkish Mathematical Society Undergraduate & Graduate Summer School, 2019, Nesin Mathematics Village
    - I took 1-week courses on Category Theory (Matteo Paganin) and Bernoulli Polynomials (Mehmet Cenkci), 2-week course on Differential Geometry (Özgür Kişisel & İlker Berktav).
13. Turkish Women in Mathematics Graduate Summer School, 17-26 June 2019, Middle East Technical University
    - In this school we discussed the first ≈80 pages of Harris' Algebraic Geometry book. Lecturer was Özgür Kişisel.
14. Middle East Technical University Math Society Workshops
15. 7th Cemal Koç Algebra Day, 27 April 2019, Bilkent University
16. Turkish Mathematical Society Undergraduate & Graduate Summer School, 2018, Nesin Mathematics Village
    - I took 1-week courses on Ring Theory (Ali Nesin), Lie Algebras (Şükrü Yalçınkaya), Manifolds & Special Holonomy (Özgür Kelekçi).
17. Nesin Mathematics Village Winter School, 2018
    - I took 2-week course on Module Theory (Ali Nesin).
18. Nesin Mathematics Village Summer School, 2017
    - This summer school is offered for high-school students. I attended this school in my last year of high-school education. I took 2-week courses on Group Theory (Ali Nesin) and Ring Theory (Salih Durhan).

Scholarships

- **Ozyegin University**, *Fellowship Package (Sept. 2021 — July 2023)*: Full tuition waiver, monthly net stipend, free dormitory room, and private health insurance.

- **Ozyegin University**, Travel grant for "Horizons in non-linear PDEs Summer School".
- **Ulm University**, Accommodation grant for "Horizons in non-linear PDEs Summer School".

Teaching Experience

- *Fall 2026*: Instructor of Record for **MATH 1030** Introduction to Quantitative Reasoning at the University of Utah (57 students).
- *Spring 2026*: Lab assistant for **MATH3140** Engineering Vector Calculus and PDEs at the University of Utah.
- *Fall 2023 and Fall 2024*: Lab assistant for **MATH2250** Differential Equations and Linear Algebra at the University of Utah.

Computer Skills

- HTML, CSS, and Bootstrap
- MATLAB and Python
- LaTeX
- Linux

Service

- Peer-reviewer: *Funkcialaj Ekvacioj.*
- (Fall 2023—) Treasurer of the SIAM Student Chapter at the University of Utah.

Languages

- Turkish (Native)
- English
- Arabic